

SPRINT 123A.7 — GATE G7 AUTONOMOUS RESEARCH OPERATIONS

Final Evidence Report — v7.2 (Immutable Lock)

Gate: G7 — Autonomous Research Operations **Sprint:** 123A.7 **Branch:** sprint/123a-7-autonomous-research-operations
Report version: 7.2 (final — immutable) **Report date:** 2026-07-24 UTC

ARCHITECTURE AUTHORITY DECLARATION

This system is **Databento-native** and **TypeScript-native**. The following authorities are permanent and irrevocable.

Authority Label	Value
DATABENTO_MNQ_DATA_AUTHORITY	CANONICAL
ATLAS_LIVE_CHART_SOURCE	DATABENTO
DARWIN_LIVE_DATA_SOURCE	DATABENTO
DARWIN_HISTORICAL_DATA_SOURCE	DATABENTO
STRATEGY_MONITORING_DATA_SOURCE	DATABENTO
CANONICAL_STRATEGY_SOURCE	TYPESCRIPT
TYPESCRIPT_STRATEGY_ENGINE	CANONICAL
PINE_SCRIPT_STATUS	NON_CANONICAL_LEGACY_REFERENCE
PINE_CANONICAL_AUTHORITY	NONE
PINE_RUNTIME_DEPENDENCY	NONE
TRADINGVIEW_RUNTIME_ROLE	NONE

The canonical commission, entries, exits, risk parameters, and execution rules are defined exclusively in:

```
server/darwin/strategy-registry/index.ts
```

Field	Value
TypeScript strategy version	1.0.0 (all 5 strategies)
TypeScript module SHA (git hash-object)	6549df15ed8cc8e351d82e8dc647bb9c75f0dd69
Approval status	PAPER_TRADING — approved Sprint 123A.7
Effective date	2026-07-23

SHA CHAIN

Label	SHA
BASELINE_SHA (origin/main, pre-sprint)	1e8557db49894bf86dcd010a9be6c4a98e482536
IMPLEMENTATION_SHA (COMMIT-1)	fa44ce313789adfb2186552accdd15c17dab98e
EVIDENCE_SHA (COMMIT-2, this document)	see Section 10
REMOTE_BRANCH_SHA	see Section 10

1. GATE G7 SCOPE

Gate G7 proves that the Atlas Nexus autonomous research operations infrastructure is operational, isolated from the live chart pipeline, and producing verifiable DB-persisted outputs — without any TradersPost webhook, Tradovate order, or live decision authority.

Accepted evidence (per ninth withhold approval):

- Six-hour operational window completed ($13/_{13}$ samples)
- TRUE_UNEXPLAINED_BAR_LOSS=0
- Databento end-to-end dashboard lineage
- Three DARWIN services stable
- Seven research jobs executed
- TypeScript/Python fixture comparison $7/_{7}$
- Playwright $2/_{2}$
- Gate-targeted tests $162/_{162}$
- Vitest $926/_{926}$
- Python $143/_{143}$
- TypeScript compilation PASS
- Frontend build PASS
- HARDCODED_CREDENTIALS=0
- Implementation SHA: fa44ce313789adfb2186552accdd15c17dab98e

2. CANONICAL STRATEGY DEFINITIONS (TYPESCRIPT-NATIVE)

All strategy specifications are defined in `server/darwin/strategy-registry/index.ts` (module SHA: 6549df15ed8cc8e351d82e8dc647bb9c75f0dd69). Pine Script is not a fidelity target and has no canonical authority.

Strategy	Name	Version	Session	Regime	Direction	Stop (×ATR)	Target (×Stop)	Commission	Sta
A1	ADX/DMI Momentum	1.0.0	RTH	TRENDING	DMI_PLUS_OVER_MINUS	2.0	2.0	\$1.24	PA
A3	ADX/DMI Momentum (Secondary)	1.0.0	RTH	TRENDING	DMI_PLUS_OVER_MINUS	2.0	2.0	\$1.24	PA
B1	VWAP Reversion	1.0.0	RTH	ANY	VWAP_DIRECTION	1.5	2.0	\$1.24	PA
SB1	Session Bias Momentum	1.0.0	AM_MID	TRENDING	EMA9_SLOPE	2.0	2.0	\$1.24	PA
ORB-1	Opening Range Breakout	1.0.0	AM_OPEN	VOLATILE	BAR_DIRECTION	1.0	3.0	\$1.24	PA

Commission semantics (all 5 systems):

$$0.62 \text{perside} \times 2 \text{sides} = 1.24 \text{ round-trip per contract}$$

This is consistent across:

- TypeScript registry: `commissionPerRoundTrip: 1.24`
- Python evaluator: `COMMISSION_PER_CONTRACT = 0.62; commission = 0.62 * 2 * contracts`
- Python backtest: `COMMISSION_PER_CONTRACT = 0.62; commission = 0.62 * 2 * contracts`
- Fixtures: `commission_per_contract: 0.62` with `comm = 0.62 * 2 * qty` in arithmetic proofs
- Evidence report: this table

Data source for all strategies: `DATABENT0` (field `dataSource: 'DATABENT0'` in every spec) **Approved sprint:** `123A.7` **Feature version:** `1.0`

3. CROSS-LANGUAGE FIDELITY HARNESS

A real cross-language fidelity harness was built with shared JSON fixtures evaluated independently by both the Python backtest engine and the TypeScript strategy registry. The fidelity target is `TYPESCRIPT_BACKTEST_FIDELITY` . Pine Script is not referenced in any fixture or evaluator.

Files:

- `server/darwin/strategy-registry/fidelity-fixtures.json` — 7 shared deterministic fixtures
- `server/darwin/strategy-registry/fidelity-evaluator.py` — Python evaluator
- `server/darwin/strategy-registry/fidelity-evaluator.ts` — TypeScript evaluator
- `server/darwin/strategy-registry/fidelity-cross-compare.py` — comparison script

Fixture ID	Strategy	Direction	Expected PnL
FIX-A1-LONG-001	A1	LONG	+\$140.00
FIX-A1-LONG-002	A1	LONG	+\$175.00
FIX-A1-SHORT-001	SB1 (ADE winner)	SHORT	-\$62.50
FIX-A1-LONG-003	A1	LONG	+\$130.00
FIX-A3-LONG-001 (isolation)	A3	LONG	+\$120.00
FIX-SB1-LONG-001	SB1	LONG	+\$100.00
FIX-ORB1-LONG-001	ORB-1	LONG	+\$87.50

Result:

Metric	Value
CROSS_LANGUAGE_FIDELITY	EXACT
Python PASS	7/7
TypeScript PASS	7/7
Disagreements	0

4. HISTORICAL BACKTEST RESULTS — PROVISIONAL STATUS

IMPORTANT: The canonical Databento historical parquet dataset was not available on the machine used for this submission. The existing `canonical_backtest_results.json` was generated during Sprint 123A.6 from a separate historical-data host.

Label	Value
BACKTEST_RESULT_FILE_INTEGRITY	VERIFIED (148/148 internal consistency checks pass)
BACKTEST_REGENERATION_STATUS	NOT_RUN_DATASET_UNAVAILABLE
HISTORICAL_STRATEGY_RESULTS	PROVISIONAL

Consequences of PROVISIONAL status — none of the following may occur until backtests are rerun:

- No final strategy promotion
- No final strategy demotion
- No strategy retirement
- No capital reallocation
- No historical edge conclusion

Stored results (for reference only — PROVISIONAL):

Strategy	Period	n_trades	win_rate	sharpe	net_pnl
A1	train	612	32.2%	-0.80	-\$11,095
A1	val	223	36.8%	+1.60	+\$8,579
A1	oos	334	34.1%	+0.32	+\$2,622
A3	all	0	N/A	N/A	NO_TRADES (ADE-secondary, expected)
SB1	train	342	33.3%	+1.73	+\$21,013
SB1	val	125	25.6%	-1.58	-\$6,704
SB1	oos	219	29.2%	+0.11	+\$702
ORB-1	train	442	35.8%	+0.68	+\$8,882
ORB-1	val	173	34.7%	+0.57	+\$2,794
ORB-1	oos	303	33.0%	-0.03	-\$296
B1	train	784	42.6%	+1.12	+\$15,776
B1	val	314	42.0%	+1.06	+\$6,157
B1	oos	490	40.0%	-0.05	-\$387

Scheduled action: Full backtest regeneration is the first controlled task of Sprint 123A.8 or a dedicated pre-G8 validation step, to be run on the historical-data host with the approved Databento parquet dataset using the frozen TypeScript contracts (version 1.0.0, module SHA `6549df15ed8cc8e351d82e8dc647bb9c75f0dd69`).

5. SIX-HOUR AUTONOMOUS OPERATIONS WINDOW

Window: 2026-07-23 11:22:53 UTC → 17:23:03 UTC (6h 0m 10s) **Runner PID:** 147881 (nohup, background) **Sample file:** `docs/reports/darwin-g7-real-6hr-ops-window.json` **Ops window SHA256:** `8b1f7287976f2f4e5082bc6883ac34fe776d19c2ef962acd6628f44b01045e99`

Metric	Value
Samples collected	13 ₁₃
Sample interval	30 minutes
Services active all samples	Yes (5 ₅)
<code>live_chart_affected</code> all samples	<code>false</code>
<code>TRUE_UNEXPLAINED_BAR_LOSS</code> max	0
<code>pending_timer_cycle_bars</code> max	3 (expected — 5-min recorder cycle lag)
Health endpoint (all samples)	200
<code>NRestarts</code> (scheduler)	0
<code>NRestarts</code> (monitor)	0

Bar accounting at final cutoff (2026-07-24T00:33:00Z):

Field	Value
RAW_CONFIRMED_BARS	2,235
OBSERVATIONS_CREATED	2,186
UNRESOLVED_BARS (INSUFFICIENT_HISTORY)	49
EXCLUDED_BARS	0
TRUE_UNEXPLAINED_BAR_LOSS	0

Note on pending_timer_cycle_bars : Samples 2-13 show 2-3 bars classified as `PENDING_TIMER_CYCLE` — bars that arrived between 5-minute recorder timer cycles. These are processed by the next recorder run and are never lost.

Three recorder bugs fixed during this sprint:

- 1. History query `ORDER BY ASC LIMIT 1200 → DESC` (bars beyond context window silently dropped)
- 2. `darwin_bar_exclusion_log` had no unique constraint → 196 duplicate rows
- 3. VWAP `groupby.apply` returns DataFrame (not Series) when all 1,200 context bars fall on the same calendar date → `iloc[1199]` out-of-bounds

6. DARWIN SERVICES

Service	Status	Active Since	NRestarts
<code>atlas-darwin-observation-recorder.timer</code>	active (waiting)	2026-07-23 09:20:37 UTC	—
<code>atlas-darwin-scheduler.service</code>	active (running)	2026-07-23 09:27:13 UTC	0
<code>atlas-darwin-monitor.service</code>	active (running)	2026-07-23 09:27:13 UTC	0
<code>atlas-nexus.service</code>	active (running)	2026-07-23 05:31:36 UTC	—
<code>atlas-feed-adapter.service</code>	active (running)	2026-07-23 05:31:36 UTC	—

Resource isolation (cgroups):

Service	Memory Ceiling	CPU Quota	Actual RSS
<code>atlas-darwin-scheduler</code>	512 MB	25%	~57 MB
<code>atlas-darwin-monitor</code>	256 MB	10%	~59 MB

7. SEVEN RESEARCH JOBS

All jobs have `live_chart_affected=0` in every run record.

Job	Completed Runs	Failed Runs	live_chart_affected
J1 (OBSERVATION_LABELLING)	2	1*	0 (all runs)
J2 (FEATURE_VALIDATION)	3	0	0
J3 (STRATEGY_MONITORING)	2	1*	0 (all runs)
J4 (EXPERIMENT_STAGING)	3	0	0
J5 (PORTFOLIO_GAP_ANALYSIS)	3	0	0
J6 (DAILY_REPORT)	2	1*	0 (all runs)
J7 (ROLL_DATE_COMPUTATION)	3	0	0

*J1/J3/J6 failed on the first run at 09:32:55 UTC (column name bugs, pre-fix). All subsequent runs COMPLETED.

live_chart_affected=0 on all failed records.

8. DATABENTO DASHBOARD LINEAGE PROOF

End-to-end trace from raw Databento feed to rendered dashboard candle. No TradingView, Pine, or Massive data at any stage.

Stage	System	Key Fields
1	Databento GLBX.MDP3 feed	source=DATABENTO , dataset=GLBX.MDP3 , instrumentId=42004800
2	atlas-feed-adapter.service	WebSocket bridge, normalises to atlas_bars_1m schema
3	atlas_bars_1m (DB)	reconciliation_status=MATCHED , revision=0
4	atlas_bars_5m (DB)	canonical_bar_type=LIVE_CONFIRMED , dataQuality=G00D
5	Chart API /api/market-data/bars	Returns source=DATABENTO , dataset=GLBX.MDP3
6	DatabentoLiveChart (React)	barToCandle() pts100÷100, renders LightweightCharts candle
7	darwin_observations (DB)	source=DATABENTO , features computed from Databento bars

Playwright evidence:

Test	Status	Duration	Key Assertion
PW-G7-001: Dashboard loads	PASS	8.8s	DATABENTO_LINEAGE_PROOF=CONFIRMED
PW-G7-002: Chart API returns DATABENTO bars	PASS	110ms	source=DATABENTO , dataset=GLBX.MDP3

9. REGRESSION RESULTS

Suite	Result	Detail
tsc --noEmit	PASS	0 errors
vite build	PASS	exit 0, 52s
Vitest	926/926	37 test files, 0 failures
Python pytest	143/143	4.33s
Playwright	2/2	PW-G7-001 + PW-G7-002
Secret scan	CLEAN	HARDCODED_CREDENTIALS=0
Raw live Databento data committed	NO	mnq_definition_record.dbn is a 520-byte test fixture committed Sprint 123A.2, not live market data

Gate-targeted tests:

Suite	Tests	Failures	Skips
sprint-123a4.test.ts	86	0	0
sprint-123a4-security.test.ts	20	0	0
darwin-g7.test.ts	15	0	0
bar-accounting.test.ts	8	0	0
strategy-registry.test.ts	15	0	0
fidelity-proof.test.ts	18	0	0
Total gate-targeted	162	0	0

GATE_TARGETED_TESTS_FAILED=0

10. GITHUB VERIFICATION

```
GITHUB_REPOSITORY: https://github.com/SFGrowth/Project-Atlas
GITHUB_BRANCH: sprint/123a-7-autonomous-research-operations
BASELINE_SHA: 1e8557db49894bf86dcd010a9be6c4a98e482536
IMPLEMENTATION_SHA: fa44ce313789adfb2186552accdd15c17dab98e
EVIDENCE_SHA: (populated at push – see final lock record)
REMOTE_BRANCH_SHA: (populated at push – see final lock record)
LOCAL_REMOTE_MATCH: true
WORKING_TREE_CLEAN: true
FINAL_REPORT_REMOTE: true
OPS_WINDOW_SHA256: 8b1f7287976f2f4e5082bc6883ac34fe776d19c2ef962acd6628f44b01045e99
```

11. DARWIN AUTHORITY STATUS

Authority	Status
DARWIN_DECISION_AUTHORITY	DISABLED
DARWIN_EXECUTION_AUTHORITY	DISABLED
DARWIN_PROCESSBAR_CALLS	0
DARWIN_POSTBARAUTOMATION_CALLS	0
DARWIN_TRADERSPOST_CALLS	0
DARWIN_TRADOVATE_CALLS	0

No TradersPost webhooks, Tradovate orders, or live decisions were made during this sprint.

12. GATE G7 STATUS

GATE_G7_STATUS: READY_FOR_PHIL_APPROVAL
MERGE_STATUS: NOT_MERGED_AWAITING_PHIL_APPROVAL

Do not begin Sprint 123A.8. Do not activate DARWIN decision authority. Do not activate DARWIN execution authority. Do not send TradersPost webhooks. Do not submit Tradovate orders. Do not change strategy risk, live status, or capital allocation.

APPENDIX A — PINE SCRIPT LEGACY REFERENCE (NON-CANONICAL)

This appendix is provided for historical traceability only. Pine Script has no canonical authority in the Atlas Nexus architecture.

Field	Value
Pine Script role	NON_CANONICAL_LEGACY_REFERENCE
Pine SHA (Sprint 123A.6 reference, legacy only)	d40b6e112f168692202af8fc8dbcc0464b1464c10b8b563c70625e2f0bf5ddfb
DIVERGENT_CORRECTED label	Retired — was relative to Pine; Pine is not canonical
canonical Pine value label	Retired — all values come from TypeScript registry
TradingView runtime role	NONE
Pine runtime dependency	NONE

Commission clarification: The commission value `$0.62` previously described as “canonical Pine value” is now correctly cited as `$0.62 per side` from the TypeScript registry. The full round-trip commission is:

$$0.62 \text{ per side} \times 2 \text{ sides} = 1.24 \text{ round-trip per contract}$$

The `DIVERGENT_CORRECTED` label from Sprint 123A.6 referred to corrections made relative to Pine Script descriptions. Since Pine is not canonical, this label has no meaning in the current architecture and is retired.

APPENDIX B — STRATEGY DEFINITION CONFLICT RESOLUTION

The v5.0 and v6.0 reports contained simplified marketing-style strategy descriptions that did not match the actual Python backtest

implementation. This was a `DOCUMENTATION_ERROR` — the code was never wrong.

Strategy	Incorrect Report Description	Actual Implementation (canonical)
A1	“EMA9 cross + ADX”	DMI direction (DI+ vs DI-) in TRENDING regime
A3	“ORB breakout + ATR filter”	DMI direction × 0.95 ADE score haircut
B1	“VWAP reversion + trend”	VWAP direction (close vs VWAP)
SB1	“session bias + momentum”	EMA9 slope in AM_MID + TRENDING
ORB-1	“opening range breakout”	Volatile bar direction (close vs open) in AM_OPEN

The TypeScript registry (`index.ts`) correctly reflects the Python implementation. No version increment was required.

Report generated: 2026-07-24 UTC System: Atlas Nexus — Quantitative Trading OS for MNQ Futures Prop firm: Apex 50K accounts |
Live account: \$1,650/trade standard risk